

PAPER CODE: BCE-O534
SEMESTER EXAMINATION- DECEMBER 2024
CLASS: B.TECH SEMESTER:5TH
SUBJECT: INTRODUCTION TO A.I.
PAPER TITLE: INTRODUCTION TO A.I.

Time: 3 hour

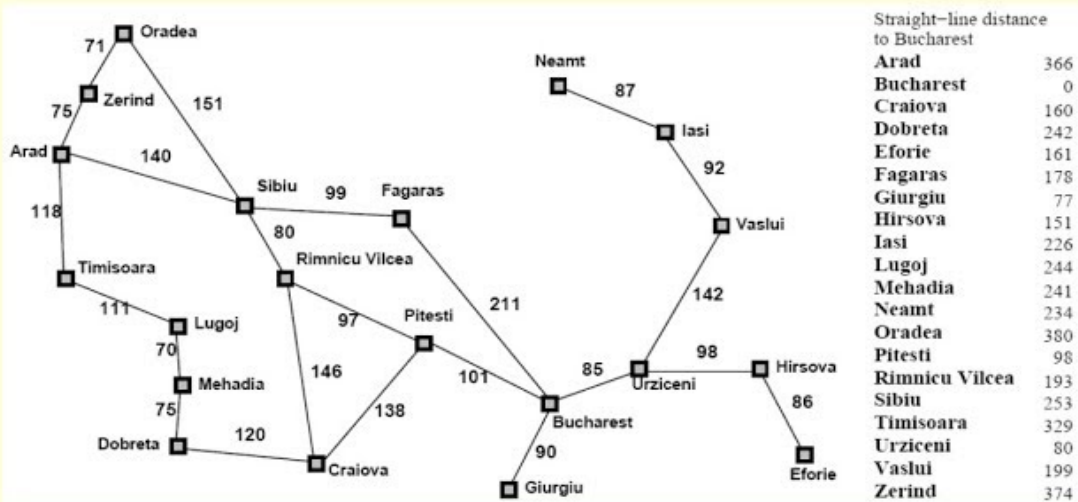
Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

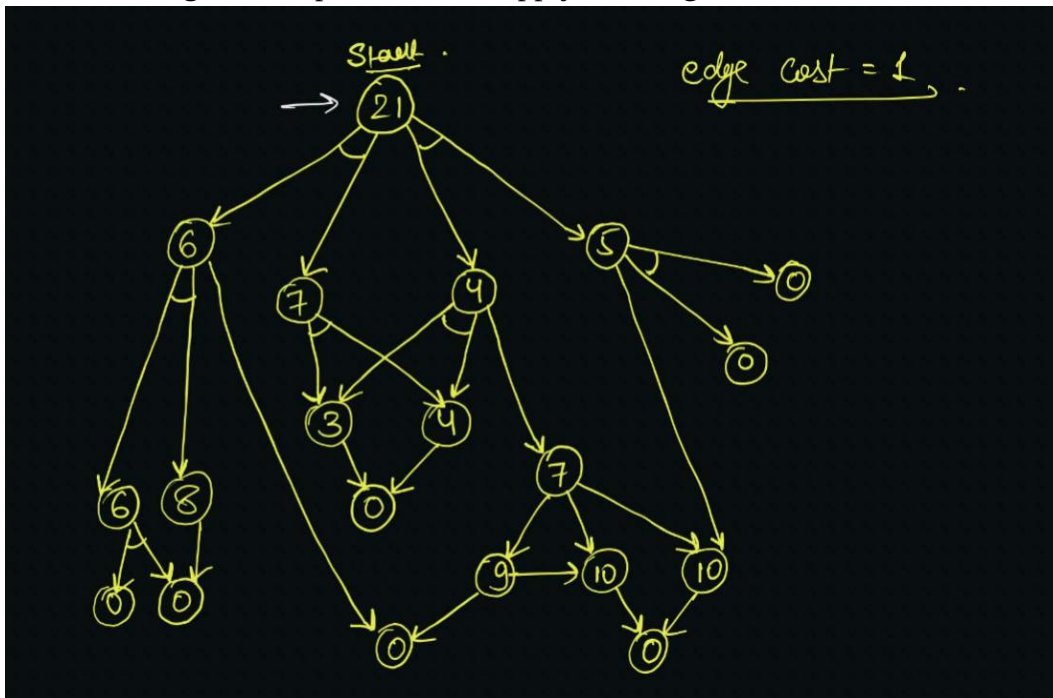
SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 X 6 = 30 Marks)			
Q-1	Explain the historical background of Artificial Intelligence and describe two applications of AI in everyday life.	1	2
Q-2	Define the concept of "search" in Artificial Intelligence and explain its significance in problem-solving.	2	1
Q-3	Discuss the limitations of local search algorithms like Hill Climbing. How can these limitations be mitigated?	2	2
Q-4	What are the differences between uniformed search strategies and informed search strategies? Provide examples of each.	2	
Q-5	There is a man who has a lion, a goat, and a bundle of grass. He has a boat too. He wants to cross a river. His boat is not very strong and therefore he can only take one thing at a time with him. How many times will he have to cross the river? (Note: if the goat and grass are left alone. the goat will eat the grass; if the lion and goat are left alone, the lion will eat the goat.)	1	4
Q-6	State the converse, contrapositive & inverse of each of them:- a) I go to the beach whenever it is a sunny summer day. b) When I stay up late, it is necessary that I sleep until noon.	4	3
Q-7	Translate these Statements into English, domain $\rightarrow$ all animals. $R(x)$: x is a rabbit. $H(x)$: x hops. $\begin{array}{ll} \text{a.) } \forall x (R(x) \rightarrow H(x)) & \text{c.) } \exists x (R(x) \rightarrow H(x)) \\ \text{b.) } \forall x (R(x) \wedge H(x)) & \text{d.) } \exists x (R(x) \wedge H(x)) \end{array}$	4	4
Q-8	Describe the difference between structured and symbolic descriptions in pattern recognition. Provide an example where each type is commonly used.	2	4
Q-9	What is the primary difference between supervised and unsupervised learning? Provide one example for each type of learning scenario.	3	2

Q-10	Explain the structure and functioning of a decision tree in machine learning.	3	1
SECTION-B (LONG ANSWER TYPE QUESTIONS) Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 X 10 = Marks)		CO	BL
Q-1	What are intelligent agents? Illustrate their structure and how they interact within an AI system.	1	3
Q-2	Consider a state space where the Start State is numbered 1 & each State k has 2 Successors: $2K$ & $2K+1$. a) Draw the portion from states 1-15 b) Suppose the goal State is 11. List out the ordering of nodes using: (i) BFS (ii) Depth limited ($l=2$, $l=3$) modes, (iii) Iterative deepening. c) Call the action of going from state $K \rightarrow 2k$ as left & $K \rightarrow 2K+1$ as right. Can you find an algorithm that outputs the Sol/path to this problem without any search?	2	6
Q-3	Describe the working mechanism of Breadth-First Search (BFS) and Depth-First Search (DFS). When is BFS preferred over DFS and vice versa?	2	3
Q-4	Compute the route to Romania, using the straight-line distance as the heuristic value. If the goal is Bucharest, and initial position is Arad, then compute the straight-line distances to Bucharest using A* Search Algorithm to find path, in the graph given below.	5	3

A* Search



Q-5 Consider the given Graph below and apply AO* algorithm on it.



Each node in the graph is assigned a heuristic value, denoted as $h(n)$, and the edge length is considered as 1.

Q-6 Which one of them is a tautology, contradiction & Contingency.

	$\begin{aligned} \text{a)} & p \rightarrow \neg p \\ \text{b)} & p \oplus (p \vee q) \\ \text{c)} & p \leftrightarrow \neg p \\ \text{d)} & (p \wedge q) \rightarrow (p \vee q) \\ \text{e)} & (q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q) \\ \text{f)} & (p \leftrightarrow q) \oplus (p \leftrightarrow \neg q) \end{aligned}$		
Q-7	Are they logically equivalent? $\begin{aligned} \text{a)} & \neg \forall x (P(x) \rightarrow Q(x)) \quad \& \quad \exists x (P(x) \wedge \neg Q(x)) \\ \text{b)} & \forall x (P(x) \rightarrow Q(x)) \quad \& \quad \forall x P(x) \rightarrow \forall x Q(x) \\ \text{c)} & \exists x P(x) \wedge \exists x Q(x) \quad \& \quad \exists x (P(x) \wedge Q(x)) \\ \text{d)} & \forall x (P(x) \leftrightarrow Q(x)) \quad \& \quad \forall x P(x) \leftrightarrow \forall x Q(x) \end{aligned}$	4	4
Q-8	Compare and contrast LISP and PROLOG as programming languages for AI applications. Which language is better suited for symbolic reasoning, and why?	6	5